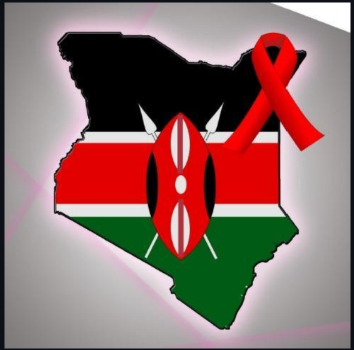
HIV CARE GAP AI

Kenya



County Risk Mapping · Individual Dropout Risk Factors · 2030 Scenario Projection

MODEL 1

County Care Gap Map

MODEL 2

Dropout Risk Factors

MODEL 3

2030 Scenario Projection

Presented by: Daniella Muli · Eve Michelle · Pheonverah Achieng' · Naomi Opiyo · Lorenah Mbogo · Dennis Nderitu

Using Machine Learning to Predict Who Kenya is Leaving Behind

What We'll Cover Today

A narrative from the problem to the solution to the impact

01 **The Problem**

Kenya's HIV crisis and why current systems fail

02 **Our Data**

NSDCC programme data + Kenya DHS 2022

03 **Model 1 : Where**

County Care Gap Map: 47 counties ranked & tiered

04 **Model 2 : Who**

Dropout risk factor analysis via logistic regression

05 **Model 3 : What Next**

2030 dual scenario projection: Do Nothing vs. Intervene

06 **The Dashboard**

Streamlit app for MOH — live, updatable, deployable

07 **Recommendations**

Three data-driven actions for the Ministry of Health

01 THE PROBLEM

Kenya's HIV crisis is accelerating, not slowing down

19,991

New HIV infections in 2024

+19% year-on-year

60%

of infections from just 10 counties

Yet no AI system ranks all 47

28+ days

missed = Treatment Interruption

IIT not modelled predictively

0

existing tools for 2030 scenarios

No county-level forecasting system

The Care Cascade — Where Kenya Loses Patients

Diagnosed → Linked to Care → On ART → **Virally Suppressed**

Dropout at any stage creates a Care Gap

OUR SOLUTION

Three ML models answering three questions the MOH has never had answers to

WHERE?

Model 1 - County Care Gap Map

KMeans clustering groups all 47 counties into Critical, High, Moderate & Low tiers using a Care Gap Index built from NSDCC raw programme data. MOH can immediately see which counties need urgent retention interventions.

KMeans (k=4) + Care Gap Index

WHO?

Model 2 - Dropout Risk Factor Analysis

Logistic Regression on 32,156 DHS 2022 individual records identifies demographic & behavioural characteristics statistically associated with HIV care dropout. Output: odds ratios with 95% CI for all 17 features.

Logistic Regression + Bootstrap CI

WHAT NEXT?

Model 3 - 2030 Dual Scenario Projection

Tier-based scenario projection shows IIT & VLS rate trajectories to 2030 under Business As Usual vs. Bridged Gap intervention in Critical & High counties. Quantifies lives retained on ART.

Scenario Projection (PEPFAR-calibrated)

Three approved, confirmed datasets from real Kenya government data

NSDCC Raw Programme Data

analytics.nsdcc.go.ke/estimates

- Adult on ART (47 counties × periods)
- HIV Testing Services (HTS)
- Viral Load Testing & Suppression (VLT)
- Interruption in Treatment (IIT) - (9 regions expanded to 47)
- PLHIV Estimates for HTS positivity rate

DHS Kenya 2022 Individual Recode

dhsprogram.com - KEIR8CDT.ZIP ⟶ individual_features.csv

- 42,300 households surveyed nationally
- 5,925 columns ⟶ reduced to 15 relevant features
- Age, sex, education, wealth, county, HIV status
- ART initiation, testing history, distance to facility
- 32,156 final records for Model 2

Data Pipeline

Extract ⟶ Clean ⟶ Merge ⟶ Feature Eng. ⟶ Train Models ⟶ Dashboard

Key Data Challenges Overcome

- IIT data region-level only (9 MOH regions) ⟶ expanded to 47 counties using equal-share distribution
- County name inconsistencies across all files ⟶ standardised via COUNTY_NAME_MAP
- VLT file single-period snapshot, no Period column ⟶ joined on county only
- Only 2025 data available ⟶ switched from Prophet to scenario-based projection
- Extreme class imbalance in dropout target: 26 of 32,156 (0.08%) ⟶ reframed to risk factor identification

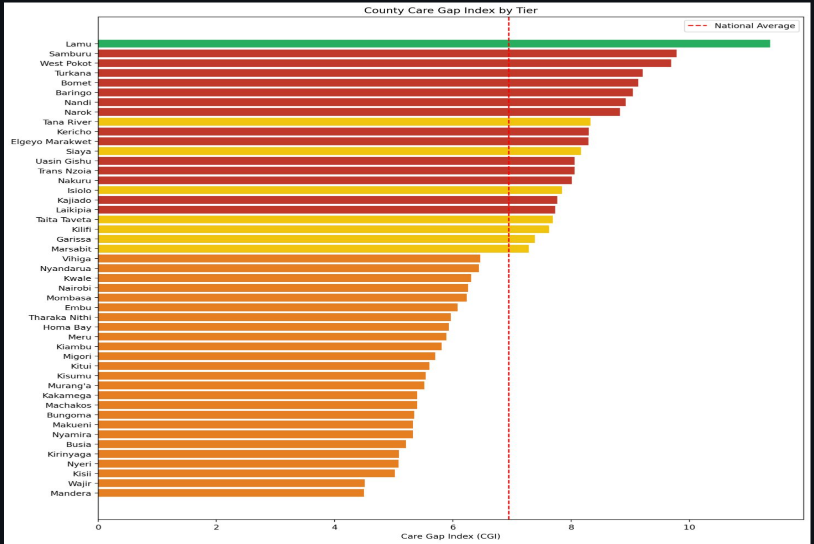
County Care Gap Map: KMeans Clustering of All 47 Counties

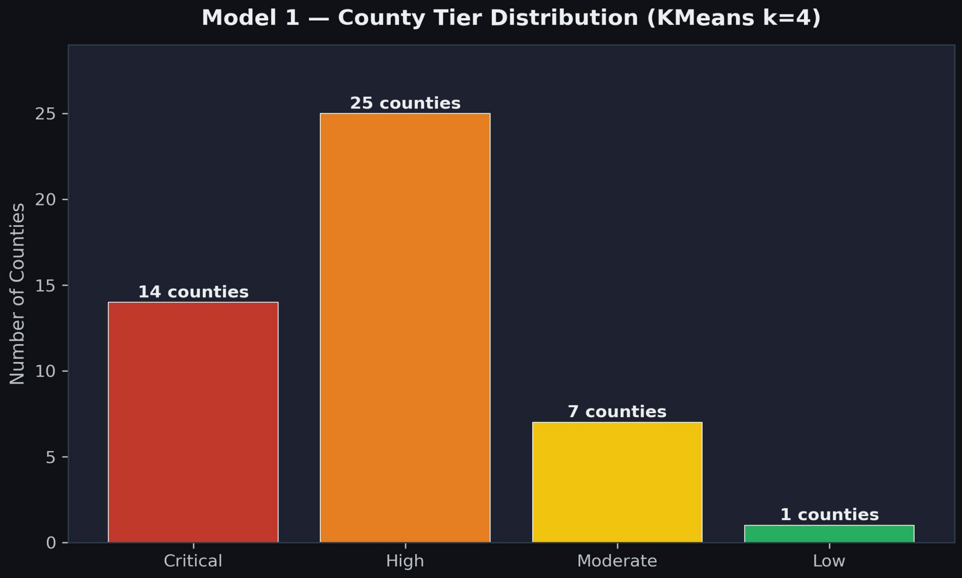
Care Gap Index Formula

CGI = (0.4 × IIT rate) + (0.4 × (1 − VLS rate)) + (0.2 × HTS positivity rate) × 100

IIT rate = treatment interruption · VLS rate = viral load suppression · Scaled 0–100

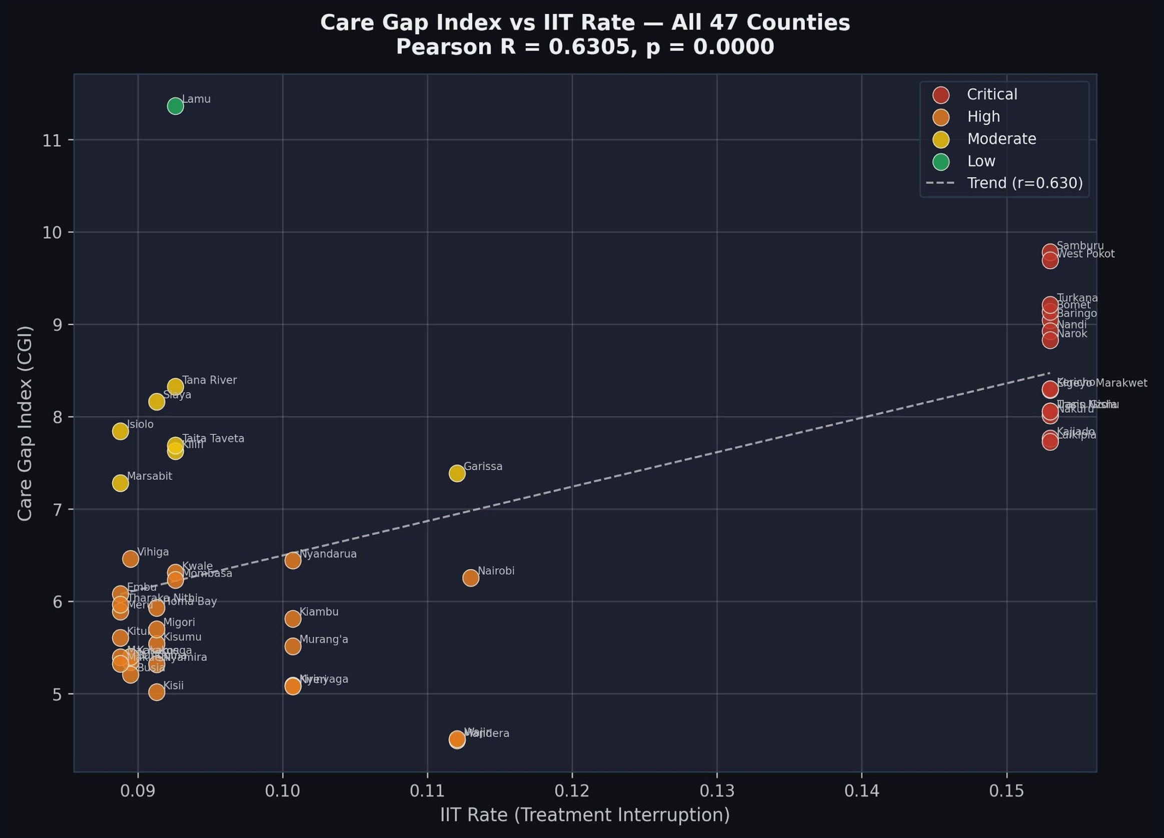
Critical	High	Moderate	Low
14 counties	25 counties	7 counties	1 county





Silhouette Score: 0.64 (k=4) — Strong cluster separation. k=3 scored 0.66 but k=4 chosen for intervention framework granularity.

Care Gap Index vs IIT Rate -> Pearson R = 0.63



Samburu & West Pokot

Highest IIT rates (>15%) AND highest CGI - rightmost Critical counties demanding immediate intervention

Lamu (Outlier)

High CGI despite low IIT rate - driven by poor VLS & HTS positivity. Unique island geography limiting facility access

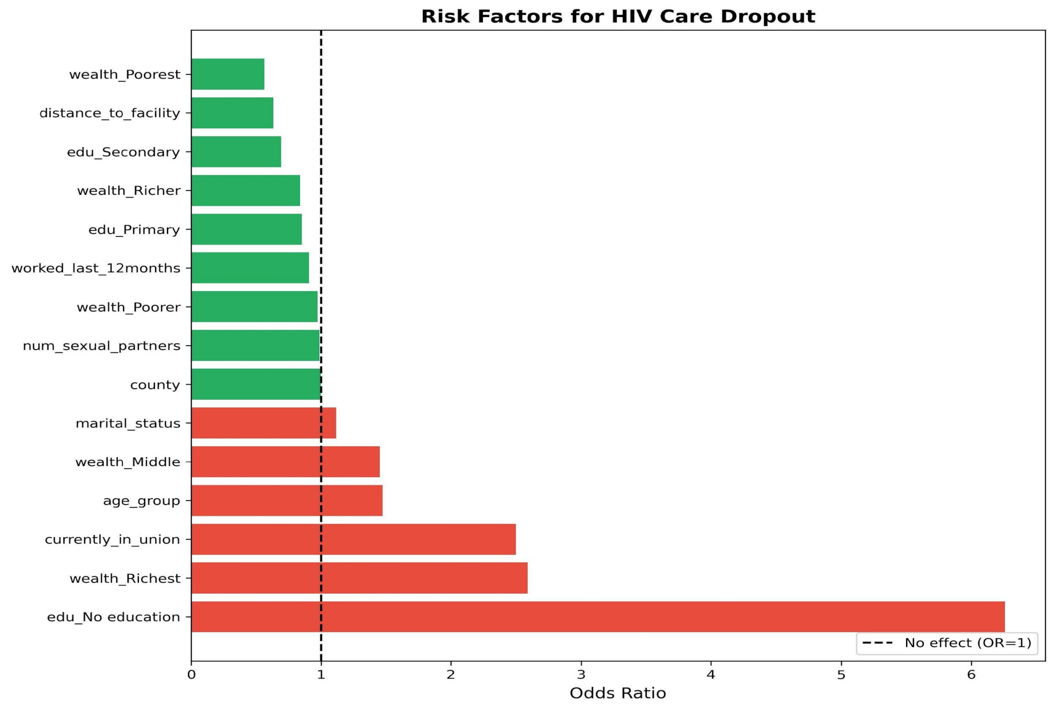
IIT Data Limitation

IIT sourced at 9 MOH regional level. Counties in the same region share the same IIT rate - documented limitation, not a bug

Pearson R = 0.63 (p < 0.0001) confirms the Care Gap Index captures real programme gaps in the data

Dropout Risk Factor Identification - Not Prediction

Important framing: With only 26 dropout cases in 32,156 records (0.08%), we deliberately reframed Model 2 from dropout prediction to dropout risk factor identification. Logistic regression odds ratios remain valid and clinically interpretable even with extreme class imbalance.



Key Findings

edu_No education OR = 6.3 ↑

Highest risk - 6× more likely to disengage from care

wealth_Richest OR = 2.6 ↑

Wealthier individuals paradoxically at higher risk

currently_in_union OR = 2.5 ↑

Marital union associated with higher dropout risk

wealth_Poorest OR = 0.5 ↓

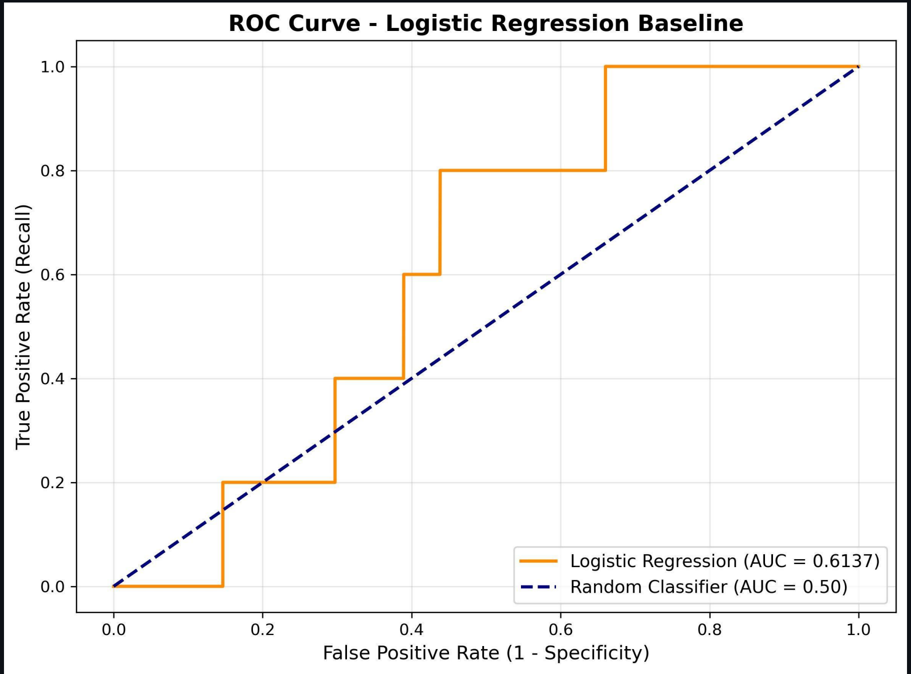
Protective - poorest may access safety-net services

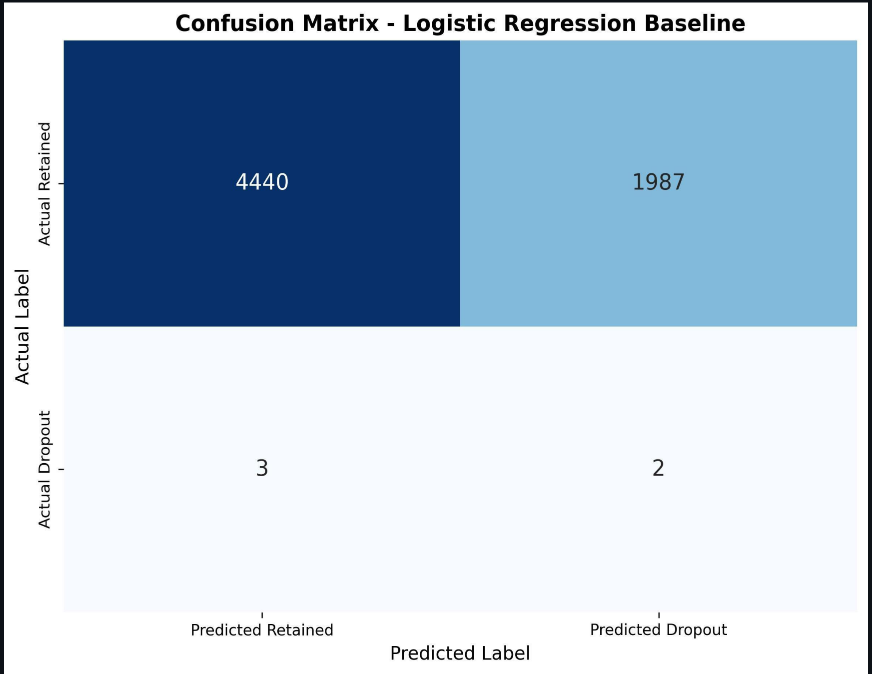
edu_Secondary OR = 0.65 ↓

Secondary education is a retention protective factor

Bootstrap 95% CI computed across 500 samples. ever_tested_hiv shows unstable OR (≈35) due to near-uniform distribution among 26 cases — documented & display-capped at OR=40.

Logistic Regression - Evaluation Results





AUC-ROC
0.614
Above random (0.5). With 26 positives, this is the correct primary metric.

Recall
0.40
40% of actual dropout cases flagged - reasonable given extreme imbalance.

F1 Score
Low
Expected with 0.08% positive rate. Odds ratios are the real deliverable.

OR Framework
Valid
Coefficients + bootstrap CIs are the clinically interpretable output.

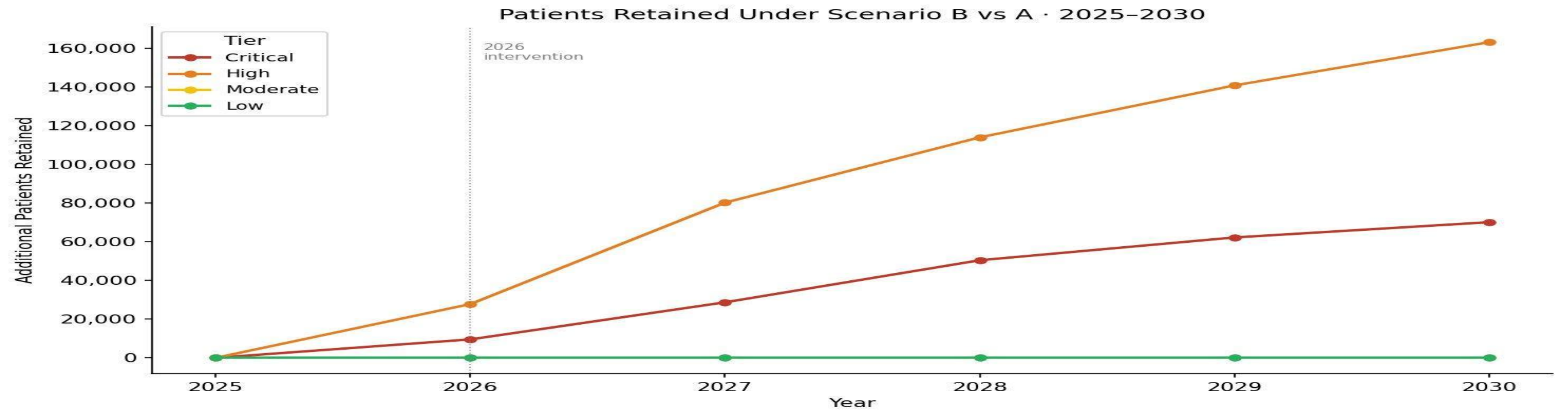
2030 Dual Scenario Projection - Do Nothing vs. We Intervene

Scenario A : Business As Usual

IIT rates held flat at 2025 baseline through 2030. No intervention. Represents what happens if Kenya does nothing.

Scenario B : Bridged Gap (Intervention)

30% IIT reduction in Critical & High counties from 2026. VLS improves at δVLS = –0.5 × δIIT (PEPFAR retention evidence). Moderate & Low tiers unchanged.



By 2030: Scenario B retains ~70,000 additional patients in Critical tier + ~163,000 in High tier - using actual adults_on_art counts as denominator, not a placeholder.

Streamlit — Three Tabs, Live on Streamlit Cloud

Tab 1

County Gap Map

- 47-county risk league table ranked by CGI
- Tier labels: Critical / High / Moderate / Low
- Folium choropleth map colour-coded by tier (stretch goal)
- CSV download for MOH policy briefs

Tab 2

Dropout Risk Factors

- Odds ratio forest plot for all 17 features
- Top risk factors: OR > 1.5 highlighted in red
- Protective factors: OR < 0.7 highlighted in green
- 95% CI bands from 500 bootstrap samples

Tab 3

2030 Forecast

- Dual scenario line charts per county tier
- IIT rate & VLS rate projections 2025–2030
- Patients retained counter - Scenario B vs A
- National headline chart across all tiers

🔄 Annual Renewal System

trigger_predictions.py --new-data-year 2026

When NSDCC publishes 2026 data, a single command re-cleans all files, re-merges, re-runs feature engineering, retrains all 3 models, logs which counties changed tier, and updates the dashboard. Kenya gains an annually renewable HIV programme monitoring system - not a one-time analysis.

MODEL EVALUATION SUMMARY

How We Know Each Model is Working

Model 1 KMeans County Clustering

Silhouette Score: 0.64

Strong cluster structure - counties well-assigned to tiers. k=3 scored 0.66 but k=4 selected for 4-tier intervention framework. Pearson R = 0.63 between CGI and IIT rate confirms index validity.

Model 2 Logistic Regression Risk Factors

AUC-ROC: 0.614 | Recall: 40%

AUC above random baseline (0.5). Primary deliverable is the odds ratio table, not classification. Bootstrap 95% CI across 500 samples validates statistical reliability of all 17 feature associations.

Model 3 2030 Scenario Projection

PEPFAR-calibrated assumptions

Scenario B validated against PEPFAR Kenya programme trend direction. $\delta VLS = -0.5 \times \delta IIT$ based on published retention evidence. Patients-retained counter uses actual adults_on_art county counts. No backtesting possible with single-year data - documented transparently.

Three Actions the Ministry of Health Can Take Now

01 Model 1 Prioritise Retention in Critical Tier Counties

Immediately target the 14 Critical tier counties identified by the Care Gap Index - specifically those with IIT rates above the national average and VLS rates below 90%. These counties represent the highest return-on-investment for reducing new infections. Samburu, West Pokot, Bomet, and Baringo show the most urgent combined signals.

ACTION: Allocate retention intervention resources to Critical tier counties first. Use the CGI league table for priority ranking.

02 Model 2 Deploy Differentiated Follow-Up by Demographic Profile

Target differentiated outreach to the demographic profiles with the highest odds ratios - specifically individuals with no formal education (OR = 6.3), those currently in union (OR = 2.5), and those in the wealthiest wealth quintile (OR = 2.6). Secondary education and poorest wealth quintile are protective factors.

ACTION: Train community health workers to prioritise the high-OR profiles. These are statistical associations - always combine with clinical judgment.

03 Model 3 Invest in Retention Now - The 2030 Numbers Justify It

Scenario B projects that reducing IIT by 30% in Critical and High tier counties from 2026 retains approximately 70,000+ additional patients in Critical counties and 163,000+ in High counties by 2030. The VLS improvement follows at $\delta VLS = -0.5 \times \delta IIT$, based on PEPFAR retention programme evidence.

ACTION: Use Scenario B projections as a quantified evidence base for donor investment proposals - PEPFAR, UNAIDS, World Bank.

What This Project Delivers And Why It Matters

Novel Contribution

No existing Kenya HIV tool combines raw NSDCC programme data processing, individual dropout risk factor analysis, and county-level scenario forecasting in one integrated system. This project is the first county-level HIV programme data AI system built entirely from raw NSDCC data - immediately extensible to future years as new data is released.

📍 County Care Gap Map

47 counties ranked and tiered by CGI - built from raw NSDCC programme data. Ready for MOH strategic planning.

👤 Risk Factor Identification

17 demographic features ranked by odds ratio with 95% CI. Tells community health workers who to prioritise for outreach.

📈 2030 Scenario Projection

Quantified evidence base for donor investment: ~233K additional patients retained by 2030 under Scenario B.

🖥️ Streamlit Dashboard

Three-tab interactive MOH tool deployed on Streamlit Cloud. Updatable annually with new NSDCC data.

🔄 Annual Renewal System

trigger_predictions.py reruns the full pipeline when 2026 data is released - Kenya's first renewable HIV AI system.

📋 CRISP-DM Notebooks

10 notebooks documenting every decision from extraction to deployment. Fully reproducible, auditable methodology.

HOW TO ACCESS THE DASHBOARD

Live on Streamlit Cloud - No Installation Required

For MOH & Stakeholders (No Setup)

1. 1. Open any browser
2. 2. Navigate to the Streamlit Cloud URL (https://hivcaregapai.streamlit.app/)
3. 3. Tab 1 ⟶ County Risk Map | Tab 2 ⟶ Dropout Risk Factors | Tab 3 ⟶ 2030 Forecast
4. 4. Download CSV reports using the built-in export buttons

For Technical Users (Local Run)

```
git clone <repo> ⟶ pip install -r requirements.txt ⟶ python main.py ⟶ streamlit run app/streamlit_app.py
```

Updating with New Data

When NSDCC releases 2026 programme data:

```
python app/trigger_predictions.py --new-data-year 2026
```

⟶ Full pipeline reruns automatically

Thank You

Kenya gains the first county-level HIV programme data AI system
built entirely from raw NSDCC data.

WHERE?

14 Critical counties identified by Care Gap Index

WHO?

edu_No education (OR=6.3) leads risk factor table

WHAT NEXT?

~233K+ patients retained by 2030 under Scenario B

Built on NSDCC Raw Programme Data · DHS Kenya 2022 · Streamlit · CRISP-DM

Stakeholders: Ministry of Health · County Directors of Health · NSDCC · NASCOP · PEPFAR · UNAIDS · World Bank

Built by: Daniella, Eve, Verah, Naomi, Lorenah, Dennis

Questions Welcome